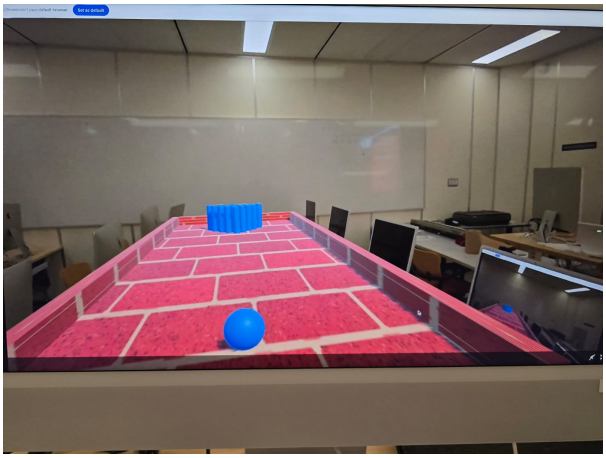
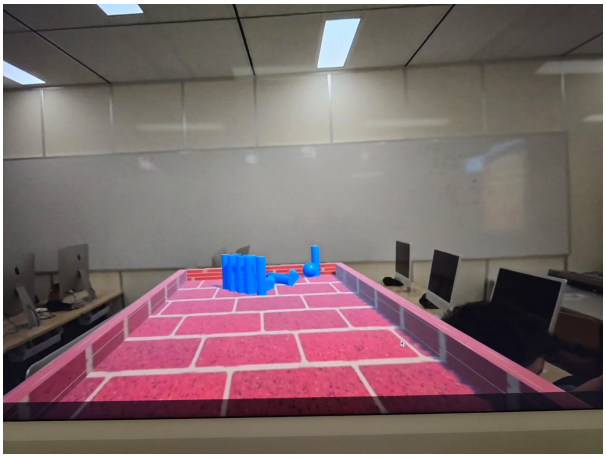
Images of Recent Work

Virtual Reality (VR)

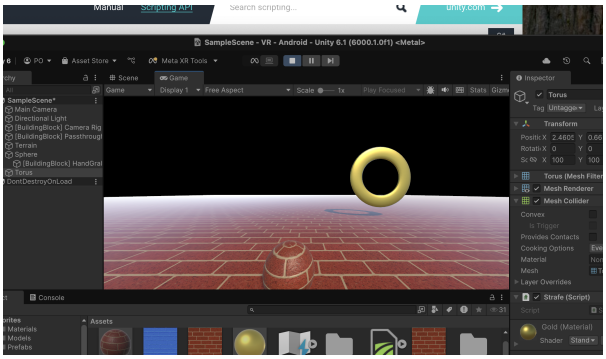


Before shot: the bowling ball is positioned and aimed toward the standing pins.



After shot: several pins have fallen, demonstrating accurate collision detection and physics simulation.

Figure 1: A collaborative student project developed as part of a small-scale Virtual Reality (VR/XR) prototype. The interactive experience features a red-bricked floor, blue pins, and a bowling ball within a physics-based simulation. The objective is for the player to achieve a strike by knocking down all pins using motion-controlled input.



Click the image to view the animation on YouTube

Scan to watch video

Figure 2: A student-developed Virtual Reality (VR) game prototype in which the player must throw a ball toward a moving vertical golden ring. The interactive experience demonstrates motion tracking, real-time physics, and dynamic target movement within an immersive 3D environment. The project explores precision-based gameplay mechanics and user engagement in spatial VR interaction design.

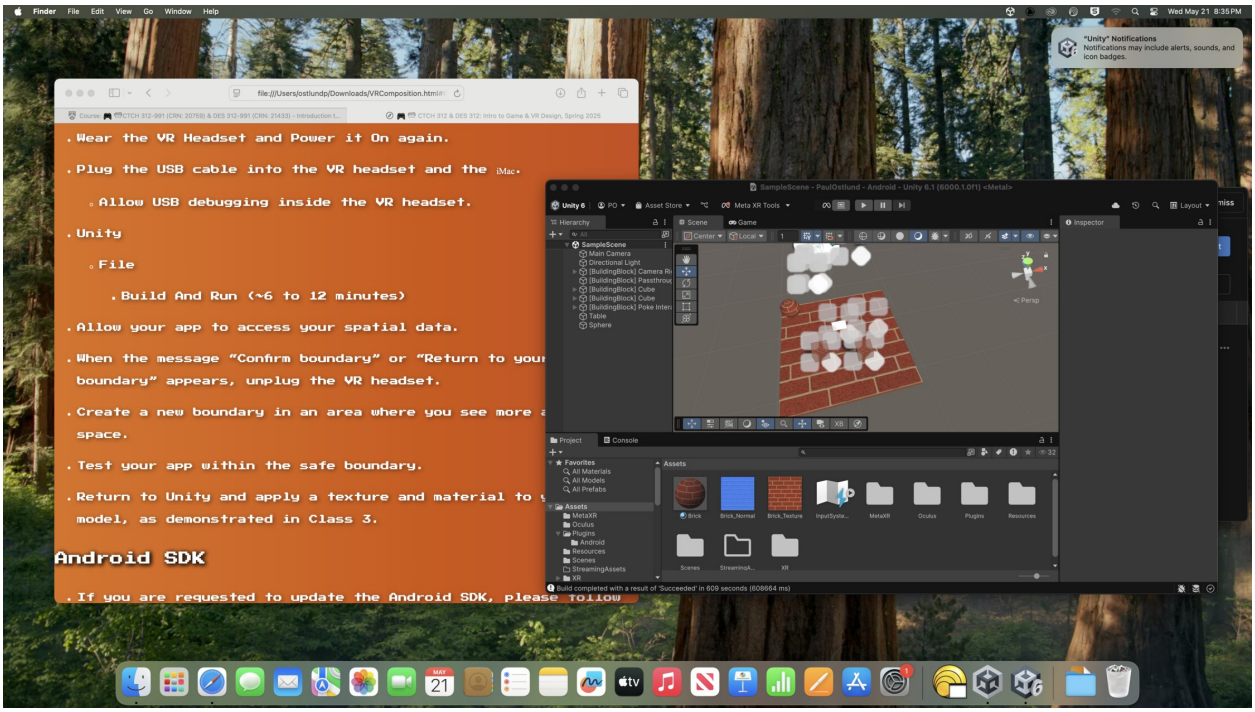


Figure 3: A student participating in the VR lab session, following step-by-step class instructions to construct a simple interactive scene. A table, a sphere, and two cubes are visible in the virtual environment, demonstrating foundational object manipulation and spatial awareness in VR design.

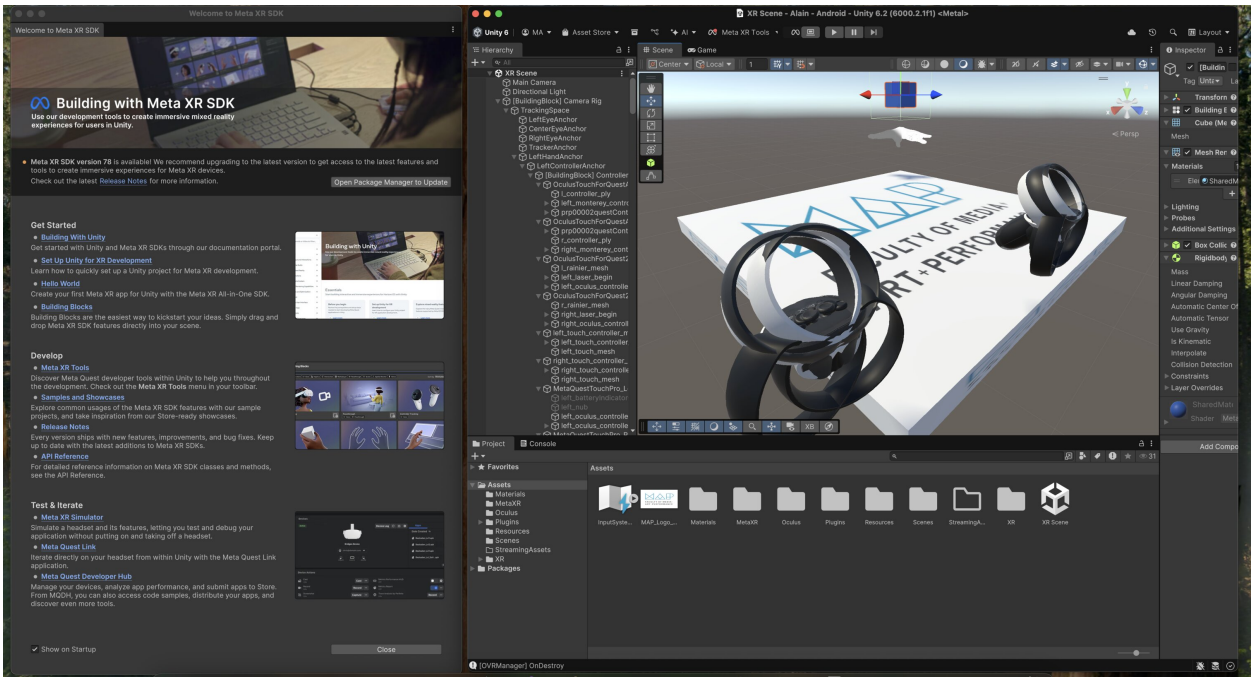


Figure 4: An instructor-created sample project built with the Meta XR SDK, demonstrating object manipulation techniques in a virtual environment. The example showcases grab, move, and release interactions with 3D objects, providing students with a reference for implementing spatial interaction mechanics in extended reality (XR) development. Project source available on GitHub (click here).



Click the image to view the playlist on YouTube

Scan to watch the playlist

Figure 5: A link to a curated YouTube playlist featuring Virtual Reality (VR) student final projects and attendee experiences from the CTCH/DES Showcase. The playlist highlights interactive scenes, object manipulation exercises, and immersive demonstrations created with Unity.